

EA-0025 - AQELYN Engineering Archive

AQELYN - EA-0025 Engineering Archive

IS-025 - AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

****Archive ID:**** EA-0025

****Implementation Specification:**** IS-025

****Component:**** AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

****Project:**** AQELYN

****System Type:**** Cyber Security Operating Environment

****Status:**** COMPLETE

****Repository Impact:**** No top-level repository structure changes

****Breaking Changes:**** None

****Engineering Phase:**** Phase 3

****Predecessor Archives:**** EA-0001 through EA-0024

****Next Specification:**** IS-026 - AQELYN Configuration Compliance & Drift Intelligence Engine

Document Control

| Field | Value |

---|---

| Document | Engineering Archive EA-0025 |

| Specification | IS-025 - AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |

| Publication Format | Markdown, PDF, HTML, ZIP |

| Source of Truth | MD/EA-0025.md |

| Archive Rule | Implementation Specification -> Engineering Archive -> Continue |

| Repository Rule | Fixed repository structure; no redesign |

| Completion State | IS-025 complete; EA-0025 generated |

1. Engineering Context

AQELYN is being built as a modular Cyber Security Operating Environment.

Completed engineering archive chain:

| Engineering Archive | Implementation Specification | Component |

---|---|---

| EA-0001 | IS-001 | AQELYN Kernel |

| EA-0002 | IS-002 | Universal Object Model |

| EA-0003 | IS-003 | AQELYN Event Bus |

- | EA-0004 | IS-004 | AQELYN Evidence Engine |
- | EA-0005 | IS-005 | AQELYN Knowledge Graph |
- | EA-0006 | IS-006 | AQELYN Trust Engine |
- | EA-0007 | IS-007 | AQELYN Mission Engine |
- | EA-0008 | IS-008 | AQELYN Workflow Engine |
- | EA-0009 | IS-009 | AQELYN Policy Engine |
- | EA-0010 | IS-010 | AQELYN Compliance & Governance Engine |
- | EA-0011 | IS-011 | AQELYN Identity & Access Governance Engine |
- | EA-0012 | IS-012 | AQELYN Asset & Configuration Governance Engine |
- | EA-0013 | IS-013 | AQELYN Risk Intelligence Engine |
- | EA-0014 | IS-014 | AQELYN Threat Intelligence Fusion Engine |
- | EA-0015 | IS-015 | AQELYN Security Operations (SOC) Engine |
- | EA-0016 | IS-016 | AQELYN Digital Forensics Engine |
- | EA-0017 | IS-017 | AQELYN Threat Detection & Analytics Engine |
- | EA-0018 | IS-018 | AQELYN Automated Response & Orchestration Engine |
- | EA-0019 | IS-019 | AQELYN Security Data Lake & Telemetry Platform |
- | EA-0020 | IS-020 | AQELYN AI Decision Intelligence Engine |
- | EA-0021 | IS-021 | AQELYN Predictive Analytics & Forecasting Engine |
- | EA-0022 | IS-022 | AQELYN Executive Intelligence & Strategic Reporting Engine |
- | EA-0023 | IS-023 | AQELYN Threat Exposure & Attack Surface Management Engine |
- | EA-0024 | IS-024 | AQELYN Vulnerability Intelligence & Prioritization Engine |
- | EA-0025 | IS-025 | AQELYN Cyber Asset Discovery & Inventory Intelligence Engine |

The fixed repository structure remains:

```
AQELYN/  
■■■■ archive/  
■■■■ blueprint/  
■■■■ docs/  
■■■■ src/  
■■■■ tests/  
■■■■ tools/  
■■■■ build/  
■■■■ releases/  
■■■■ scripts/  
■■■■ assets/  
■■■■ examples/  
■■■■ plugins/  
■■■■ sdk/  
■■■■ api/  
■■■■ README.md
```

2. IS-025 Specification Identity

```
Specification ID: IS-025  
Name: AQELYN Cyber Asset Discovery & Inventory Intelligence Engine  
Engineering Archive Target: EA-0025  
Project: AQELYN  
System Type: Cyber Security Operating Environment  
Status: Complete  
Predecessor: IS-024 - AQELYN Vulnerability Intelligence & Prioritization Engine
```

3. Purpose

The AQELYN Cyber Asset Discovery & Inventory Intelligence Engine provides continuous enterprise-wide discovery, classification, inventory, enrichment, ownership mapping, lifecycle tracking, and governance of cyber assets across

on-premises, cloud, hybrid, edge, IoT, OT, SaaS, and container environments.

The engine establishes the authoritative cyber asset inventory for AQELYN and supplies trusted asset intelligence to risk management, threat detection, vulnerability intelligence, exposure management, compliance, executive reporting, and AI-driven decision support.

It answers:

What assets exist across the enterprise?
Who owns each asset?
Which assets are unmanaged?
Which assets are mission critical?
How are assets related?
What assets have changed?
Which assets require governance?
Can every asset be explained and audited?

4. Mission

The engine shall provide:

Continuous asset discovery
Enterprise asset inventory
Asset classification
Ownership mapping
Asset lifecycle management
Relationship discovery
Configuration enrichment
Mission-aware asset intelligence
Executive asset reporting
Compliance support
Continuous inventory validation
Policy-governed asset governance

5. Scope

5.1 In Scope

Servers
Endpoints
Cloud resources
Containers
Virtual machines
Network devices
Applications
Databases
IoT devices
OT devices
SaaS assets
Identity-linked assets

5.2 Out of Scope

Software development lifecycle
Patch deployment
License management
Hardware procurement
Financial asset accounting

6. Dependencies

IS-025 depends on:

IS-001 AQELYN Kernel
 IS-002 Universal Object Model
 IS-003 AQELYN Event Bus
 IS-004 AQELYN Evidence Engine
 IS-005 AQELYN Knowledge Graph
 IS-006 AQELYN Trust Engine
 IS-007 AQELYN Mission Engine
 IS-008 AQELYN Workflow Engine
 IS-009 AQELYN Policy Engine
 IS-010 AQELYN Compliance & Governance Engine
 IS-011 Identity & Access Governance Engine
 IS-012 Asset & Configuration Governance Engine
 IS-013 Risk Intelligence Engine
 IS-014 Threat Intelligence Fusion Engine
 IS-015 Security Operations Engine
 IS-016 Digital Forensics Engine
 IS-017 Threat Detection & Analytics Engine
 IS-018 Automated Response & Orchestration Engine
 IS-019 Security Data Lake & Telemetry Platform
 IS-020 AI Decision Intelligence Engine
 IS-021 Predictive Analytics & Forecasting Engine
 IS-022 Executive Intelligence & Strategic Reporting Engine
 IS-023 Threat Exposure & Attack Surface Management Engine
 IS-024 Vulnerability Intelligence & Prioritization Engine

7. High-Level Architecture

AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

-
- Discovery Engine
- Asset Inventory Engine
- Asset Classification Engine
- Ownership Intelligence Engine
- Relationship Discovery Engine
- Lifecycle Management Engine
- Configuration Enrichment Engine
- Knowledge Graph Connector
- Security Data Lake Connector
- AI Decision Connector
- Executive Reporting Connector
- Event Publisher

8. Functional Requirements

FR-025-001 - Continuous Discovery

The engine shall continuously discover:

- Network-connected assets
- Cloud assets
- Virtual infrastructure
- Containers
- Applications
- Databases
- Identity-linked resources
- External assets

FR-025-002 - Asset Inventory

Maintain an authoritative inventory containing:

- Unique asset identifiers
- Classification
- Ownership
- Lifecycle state
- Mission criticality
- Business criticality
- Relationships

Configuration state

FR-025-003 - Asset Intelligence

Generate:

Asset health
Asset criticality
Mission alignment
Risk contribution
Governance status
Executive summaries

FR-025-004 - Explainable Asset Intelligence

Every asset record shall include:

Evidence references
Confidence indicators
Discovery source
Ownership rationale
Relationship lineage
Lifecycle history

FR-025-005 - Governance

Support:

Approval workflows
Policy validation
Version control
Auditability
Executive review

FR-025-006 - Event Publication

Publish standardized events:

asset.discovered
asset.updated
asset.classified
asset.relationship.updated
asset.lifecycle.changed
asset.retired

9. Non-Functional Requirements

The engine shall provide:

Continuous discovery
Enterprise scalability
Low-latency inventory updates
Explainability
Auditability
Repository stability
Backward compatibility

10. Core Asset Lifecycle

Asset Discovery
↓
Classification

```

↓
Ownership Assignment
↓
Relationship Discovery
↓
Inventory Update
↓
Policy Validation
↓
Continuous Monitoring

```

11. Internal Component Architecture

The AQELYN Cyber Asset Discovery & Inventory Intelligence Engine is implemented as a modular asset intelligence platform integrated with the AQELYN Kernel, Knowledge Graph, Security Data Lake, Identity Governance Engine, Risk Intelligence Engine, Threat Intelligence Fusion Engine, Vulnerability Intelligence Engine, Attack Surface Management Engine, AI Decision Intelligence Engine, and Executive Intelligence Engine.

AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

```

■
■■■ Discovery Engine
■■■ Asset Inventory Engine
■■■ Asset Classification Engine
■■■ Ownership Intelligence Engine
■■■ Relationship Discovery Engine
■■■ Lifecycle Management Engine
■■■ Configuration Enrichment Engine
■■■ Knowledge Graph Connector
■■■ Security Data Lake Connector
■■■ AI Decision Connector
■■■ Executive Reporting Connector
■■■ Event Publisher

```

12. Component Specifications

12.1 Discovery Engine

Capabilities:

```

Network discovery
Cloud discovery
Container discovery
SaaS discovery
Identity-linked discovery
External asset discovery

```

12.2 Asset Inventory Engine

Supports:

```

Unique asset registry
Asset version history
Inventory synchronization
Duplicate detection
Inventory validation

```

12.3 Asset Classification Engine

Produces:

```

Asset type
Criticality
Business function

```

Mission alignment
Regulatory classification

12.4 Ownership Intelligence Engine

Produces:

Business owner
Technical owner
Custodian
Support group
Accountability chain

12.5 Relationship Discovery Engine

Supports:

Application dependencies
Network relationships
Identity relationships
Service dependencies
Infrastructure topology

12.6 Lifecycle Management Engine

Produces:

Provisioned
Active
Modified
Retired
Archived

12.7 Configuration Enrichment Engine

Supports:

Operating system
Installed software
Cloud metadata
Configuration state
Security posture

13. Universal Object Model Extensions

13.1 AssetRecord

AssetRecord:
asset_id
asset_type
owner
lifecycle_state

13.2 AssetRelationship

AssetRelationship:
relationship_id
source_asset
target_asset
relationship_type

13.3 AssetClassification

```
AssetClassification:  
classification_id  
business_criticality  
mission_alignment  
regulatory_classification
```

13.4 AssetLifecycle

```
AssetLifecycle:  
lifecycle_id  
current_state  
state_history  
last_updated
```

14. Knowledge Graph Integration

Relationships:

```
Asset  
↓  
owned_by  
↓  
Identity  
  
Asset  
↓  
supports  
↓  
Mission  
  
Asset  
↓  
connected_to  
↓  
Asset  
  
Asset  
↓  
contains  
↓  
Configuration
```

15. Event Bus Integration

15.1 Asset Events

```
asset.discovered  
asset.updated  
asset.retired
```

15.2 Classification Events

```
asset.classified  
asset.reclassified
```

15.3 Relationship Events

```
asset.relationship.created  
asset.relationship.updated  
asset.relationship.removed
```

15.4 Lifecycle Events

```
asset.lifecycle.changed  
asset.inventory.validated
```

16. Security Data Lake Integration

Consumes:

- Discovery telemetry
- Configuration history
- Inventory history
- Network telemetry
- Cloud telemetry

17. AI Decision Intelligence Integration

Consumes:

- Asset confidence scores
- Ownership recommendations
- Inventory quality metrics
- Relationship analysis
- Learning insights

18. Risk & Vulnerability Integration

Consumes:

- Risk scores
- Vulnerability intelligence
- Exposure intelligence
- Mission impact
- Business criticality

19. Compliance Integration

Supports:

- Asset governance
- Regulatory reporting
- Policy validation
- Audit evidence
- Configuration compliance

20. Public APIs

20.1 Asset API

- GET /assets
- POST /assets
- GET /assets/{id}

20.2 Discovery API

- POST /discovery/start
- GET /discovery/status

20.3 Relationship API

```
GET /relationships
POST /relationships
```

20.4 Inventory API

```
GET /inventory-summary
GET /asset-health
```

21. Repository Impact

Implementation shall use the approved repository structure.

```
AQELYN/
■■■ src/
■ ■■■ asset_inventory/
■■■ tests/
■ ■■■ asset_inventory/
■■■ docs/
■ ■■■ asset_inventory/
■■■ api/
■ ■■■ asset_inventory/
■■■ archive/
```

No top-level repository modifications are permitted.

22. Security Architecture

The AQELYN Cyber Asset Discovery & Inventory Intelligence Engine is the trusted subsystem responsible for continuously discovering, validating, classifying, enriching, governing, and maintaining the authoritative enterprise cyber asset inventory.

Every asset record shall be:

```
Explainable
Evidence-backed
Policy-governed
Mission-aware
Risk-aware
Ownership verified
Fully auditable
Continuously validated
```

22.1 Security Principles

```
Zero Trust
Defense in Depth
Least Privilege
Continuous Discovery
Secure by Design
Policy Enforcement
Explainable Asset Intelligence
Continuous Validation
```

22.2 Authorization Model

Supported operational roles:

```
Asset Administrator
Configuration Manager
SOC Analyst
Cloud Administrator
Infrastructure Administrator
Mission Owner
```

Compliance Officer
Executive Reviewer

All asset modifications shall be governed through the AQELYN Policy Engine.

22.3 Asset Integrity

Asset records shall maintain:

Unique asset identifier
Discovery source
Ownership metadata
Classification metadata
Configuration metadata
Relationship lineage
Lifecycle state
Audit trail

Asset history shall be append-only.

22.4 Inventory Integrity

Inventory records shall maintain:

Source lineage
Duplicate resolution state
Validation status
Confidence score
Last validation timestamp
Policy references
Evidence references
Audit record

No inventory record shall be considered authoritative without discovery source and validation metadata.

23. Asset Lifecycle

23.1 Discovery Lifecycle

Asset Discovery
↓
Classification
↓
Ownership Assignment
↓
Relationship Discovery
↓
Inventory Registration
↓
Continuous Validation

23.2 Operational Lifecycle

Provisioned
↓
Operational
↓
Modified
↓
Validated
↓
Retired
↓
Archived

23.3 Audit Lifecycle

```
Asset Created
↓
Evidence Linked
↓
Policy Validated
↓
Audit Stored
```

24. Continuous Asset Operations

The engine continuously evaluates:

```
Asset inventory
Configuration changes
Ownership changes
Relationship updates
Lifecycle transitions
Cloud resources
Container environments
Inventory integrity
```

25. Performance Requirements

The engine shall support:

```
Continuous discovery
Low-latency inventory updates
Enterprise-scale inventories
Concurrent asset synchronization
High availability
Continuous operation
```

26. Scalability Requirements

The engine shall scale to support:

```
Global enterprises
Hybrid infrastructure
Multi-cloud environments
Millions of managed assets
Distributed discovery
Long-term inventory history
```

27. Audit Requirements

Every asset operation shall generate immutable audit records.

Audit events include:

```
Asset discovered
Asset updated
Classification changed
Ownership updated
Lifecycle changed
Asset retired
```

28. Failure Handling

28.1 Discovery Failure

Discovery retried
Failure recorded
Administrator notified

28.2 Inventory Failure

Inventory synchronization retried
Previous inventory retained
Audit generated

28.3 Classification Failure

Classification recalculated
Manual review initiated
Audit recorded

28.4 Policy Failure

Asset update blocked
Policy violation recorded
Manual approval required

29. Testing Strategy

29.1 Unit Testing

Validate:

Discovery Engine
Asset Inventory Engine
Asset Classification Engine
Ownership Intelligence Engine
Relationship Discovery Engine
Lifecycle Management Engine
Configuration Enrichment Engine

29.2 Integration Testing

Verify interaction with:

Kernel
Knowledge Graph
Security Data Lake
Identity Governance Engine
Risk Intelligence Engine
Threat Intelligence Engine
Attack Surface Management Engine
Vulnerability Intelligence Engine
AI Decision Engine
Executive Reporting Engine

29.3 System Testing

Validate:

Asset discovery
Inventory synchronization
Ownership assignment
Relationship discovery

Configuration enrichment
Executive asset reporting
Audit generation

29.4 Security Testing

Verify:

Authorization
Policy enforcement
Explainability
Audit logging
Inventory integrity

29.5 Regression Testing

Verify IS-001 through IS-024 remain unaffected.

30. Acceptance Criteria

IS-025 is complete when:

Discovery Engine implemented
Asset Inventory Engine implemented
Asset Classification Engine implemented
Ownership Intelligence Engine implemented
Relationship Discovery Engine implemented
Repository unchanged
Testing documented

31. Repository Validation

Repository structure remains unchanged.

```
AQELYN/  
■■■ src/asset_inventory/  
■■■ tests/asset_inventory/  
■■■ docs/asset_inventory/  
■■■ api/asset_inventory/  
■■■ archive/
```

No top-level repository modifications are permitted.

32. Engineering Summary

IS-025 introduces the AQELYN Cyber Asset Discovery & Inventory Intelligence Engine, providing enterprise-scale cyber asset discovery, inventory governance, ownership intelligence, lifecycle management, relationship discovery, configuration enrichment, and explainable asset intelligence.

Major capabilities include:

Continuous Asset Discovery
Enterprise Asset Inventory
Asset Classification
Ownership Intelligence
Relationship Discovery
Lifecycle Management
Configuration Enrichment

Mission-Aware Asset Intelligence
Executive Asset Reporting
Policy-Governed Asset Governance

The engine integrates with all previously completed AQELYN engines while preserving repository stability, modularity, and backward compatibility.

33. Specification Status

Specification ID : IS-025
Title : AQELYN Cyber Asset Discovery & Inventory Intelligence Engine
Status : COMPLETE
Engineering Archive : READY FOR GENERATION
Next Artifact : EA-0025

34. EA-0025 Engineering Objective

The objective of IS-025 was to introduce a dedicated Cyber Asset Discovery & Inventory Intelligence Engine that enables AQELYN to maintain a trusted, continuously validated, evidence-backed, policy-governed inventory of enterprise cyber assets.

The engine extends AQELYN from vulnerability prioritization into authoritative asset intelligence and inventory governance.

35. EA-0025 Engineering Summary

The implementation specification defines a modular subsystem responsible for:

Continuous asset discovery
Authoritative asset inventory
Asset classification
Ownership intelligence
Relationship discovery
Lifecycle management
Configuration enrichment
Knowledge Graph integration
Security Data Lake integration
Risk and vulnerability integration
AI Decision integration
Executive reporting integration
Event publishing

36. Major Engineering Decisions

36.1 Decision 1 - Dedicated Asset Discovery & Inventory Engine

Asset discovery and inventory intelligence are implemented as a standalone engine rather than embedded in Asset Governance, Attack Surface Management, or Vulnerability Intelligence.

Rationale:

Clear separation of inventory authority from exposure and vulnerability analysis.
Independent lifecycle and governance.
Better support for authoritative asset records and ownership mapping.
Improved traceability for asset lifecycle changes.

36.2 Decision 2 - Asset Inventory Is Evidence-Backed

Every asset record must reference discovery source, evidence, relationship lineage, confidence, and lifecycle history.

Benefits:

Inventory becomes auditable.
Ownership decisions become defensible.
Risk, vulnerability, and exposure engines receive trusted context.

36.3 Decision 3 - Asset Relationships as First-Class Objects

Asset relationships are modeled as Universal Object Model extensions.

Benefits:

Dependencies become queryable.
Mission support paths can be analyzed.
Knowledge Graph can link assets, identities, missions, risks, vulnerabilities, and exposures.

36.4 Decision 4 - Event-Driven Inventory Lifecycle

Asset discovery, classification, relationship, lifecycle, and retirement events are published through the AQELYN Event Bus.

Examples include:

```
asset.discovered
asset.updated
asset.classified
asset.reclassified
asset.relationship.created
asset.relationship.updated
asset.relationship.removed
asset.lifecycle.changed
asset.inventory.validated
asset.retired
```

36.5 Decision 5 - Universal Object Model Extension

New domain objects introduced include:

```
AssetRecord
AssetRelationship
AssetClassification
AssetLifecycle
```

37. Architectural Integration Summary

| Engine | Integration |

|--|--|

| IS-001 Kernel | Runtime lifecycle and service registration |

| IS-002 Universal Object Model | Asset, relationship, classification, lifecycle objects |

| IS-003 Event Bus | Asset, classification, relationship, lifecycle events |

| IS-004 Evidence Engine | Evidence references and asset support |

| IS-005 Knowledge Graph | Asset, identity, mission, configuration relationships |

| IS-006 Trust Engine | Data confidence and asset trust |

| IS-007 Mission Engine | Mission-aware asset criticality |

| IS-008 Workflow Engine | Asset review and governance workflows |

IS-009 Policy Engine	Asset governance, validation, and modification policies
IS-010 Compliance Engine	Asset governance and regulatory reporting
IS-011 Identity Governance Engine	Identity-linked assets and ownership context
IS-013 Risk Intelligence Engine	Risk scores and business impact
IS-014 Threat Intelligence Engine	Threat context related to asset categories
IS-019 Security Data Lake	Discovery telemetry and configuration history
IS-020 AI Decision Engine	Ownership recommendations and confidence scores
IS-022 Executive Reporting Engine	Executive asset reporting
IS-023 Attack Surface Management Engine	Exposure and attack surface context
IS-024 Vulnerability Intelligence Engine	Vulnerability and remediation context

38. Repository Impact Summary

Repository structure remains unchanged.

Implementation is expected within existing project directories, including:

```
AQELYN/  
■■■ src/asset_inventory/  
■■■ tests/asset_inventory/  
■■■ api/asset_inventory/  
■■■ docs/asset_inventory/  
■■■ archive/
```

No top-level directories were added, removed, or renamed.

39. Security Impact Summary

The specification introduces asset-inventory-specific security controls:

```
Policy-governed asset modifications  
Evidence-backed asset records  
Continuous inventory validation  
Asset ownership verification  
Relationship lineage  
Lifecycle audit trail  
Inventory confidence scoring  
Configuration enrichment traceability  
Role-authorized asset administration
```

No reduction in the security posture of existing components was identified.

40. Capabilities Added

The engine enables AQELYN to support:

```
Continuous asset discovery  
Enterprise asset inventory  
Asset classification  
Ownership mapping  
Asset lifecycle management  
Relationship discovery  
Configuration enrichment  
Mission-aware asset intelligence  
Executive asset reporting  
Compliance support  
Continuous inventory validation
```

Policy-governed asset governance

41. Risks Identified

| Risk | Mitigation |

|---|---|

| Discovery blind spots | Multiple discovery sources and continuous discovery |

| Duplicate assets | Duplicate detection and inventory validation |

| Incorrect ownership | Ownership Intelligence Engine and manual review |

| Stale inventory | Continuous validation and lifecycle events |

| Unauthorized asset changes | Policy enforcement and role authorization |

| Weak relationship mapping | Relationship lineage and confidence indicators |

| Incomplete configuration data | Configuration enrichment and source lineage |

| Poor mission alignment | Mission Engine integration |

No critical architectural risks were identified that require redesign.

42. Verification Summary

The specification defines verification for:

Unit testing
Integration testing
System testing
Security testing
Regression testing

Acceptance criteria cover discovery engine, asset inventory engine, classification engine, ownership intelligence engine, relationship discovery engine, repository validation, and testing documentation.

43. Engineering Principles Confirmed

The implementation complies with established AQELYN principles:

Modular architecture
Event-driven communication
Immutable evidence references
Traceability
Explainability
Security by design
Repository stability
Backward compatibility
Governance-before-archive discipline

44. Dependencies

Required:

EA-0001 through EA-0024
IS-001 through IS-025

Enables:

IS-026 and subsequent configuration compliance, drift intelligence, identity threat detection, cloud posture, SaaS posture, supply chain, and cyber resilience components

45. Completion Record

Engineering Archive : EA-0025
 Implementation Specification : IS-025
 Title : AQELYN Cyber Asset Discovery & Inventory Intelligence Engine
 Engineering Status : COMPLETE
 Repository Status : UNCHANGED
 Architecture Status : EXTENDED
 Backward Compatibility : MAINTAINED

46. Archive Index Update

EA-0001 IS-001 AQELYN Kernel
 EA-0002 IS-002 Universal Object Model
 EA-0003 IS-003 AQELYN Event Bus
 EA-0004 IS-004 AQELYN Evidence Engine
 EA-0005 IS-005 AQELYN Knowledge Graph
 EA-0006 IS-006 AQELYN Trust Engine
 EA-0007 IS-007 AQELYN Mission Engine
 EA-0008 IS-008 AQELYN Workflow Engine
 EA-0009 IS-009 AQELYN Policy Engine
 EA-0010 IS-010 AQELYN Compliance & Governance Engine
 EA-0011 IS-011 AQELYN Identity & Access Governance Engine
 EA-0012 IS-012 AQELYN Asset & Configuration Governance Engine
 EA-0013 IS-013 AQELYN Risk Intelligence Engine
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 EA-0018 IS-018 AQELYN Automated Response & Orchestration Engine
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 EA-0020 IS-020 AQELYN AI Decision Intelligence Engine
 EA-0021 IS-021 AQELYN Predictive Analytics & Forecasting Engine
 EA-0022 IS-022 AQELYN Executive Intelligence & Strategic Reporting Engine
 EA-0023 IS-023 AQELYN Threat Exposure & Attack Surface Management Engine
 EA-0024 IS-024 AQELYN Vulnerability Intelligence & Prioritization Engine
 EA-0025 IS-025 AQELYN Cyber Asset Discovery & Inventory Intelligence Engine

47. Engineering Phase Status

Completed Engineering Archives : EA-0001 through EA-0025

Current Status:
 EA-0025 COMPLETE

Next Implementation Specification:
 IS-026 - AQELYN Configuration Compliance & Drift Intelligence Engine

48. Engineering Archive Publication Standard

EA-0025 follows the AQELYN Engineering Archive Publication Standard.

Required package structure:

EA-xxxx/
 ■
 ■■■ diagrams/
 ■ ■■■ Architecture.svg
 ■ ■■■ Component.svg
 ■ ■■■ Workflow.svg

```
■ ■■■ EventFlow.svg
■ ■■■ Integration.svg
■
■ ■■■ examples/
■ ■■■ example_*.md
■
■ ■■■ HTML/
■ ■■■ index.html
■ ■■■ styles.css
■ ■■■ assets/
■
■ ■■■ images/
■
■ ■■■ journal/
■ ■■■ Engineering_Journal.md
■
■ ■■■ manifest/
■ ■■■ manifest.json
■
■ ■■■ MD/
■ ■■■ EA-xxxx.md
■
■ ■■■ PDF/
■ ■■■ EA-xxxx.pdf
■
■ ■■■ requirements/
■ ■■■ Requirements_Matrix.md
■
■ ■■■ traceability/
■ ■■■ Traceability_Matrix.md
■
■ ■■■ README.md
```

The master Markdown is the source of truth. The PDF and HTML are generated from the same master Markdown and must not omit sections.

49. Requirements Matrix

Requirement ID Requirement Evidence in Archive Status			
--- --- --- ---			
FR-025-001	Continuously discover assets	Sections 8, 12, 23	Complete
FR-025-002	Maintain authoritative inventory	Sections 8, 12, 22	Complete
FR-025-003	Generate asset intelligence	Sections 8, 12, 18	Complete
FR-025-004	Provide explainable asset intelligence	Sections 8, 22, 36	Complete
FR-025-005	Support governance	Sections 8, 22, 23	Complete
FR-025-006	Publish asset events	Sections 8, 15, 36	Complete
NFR-025-001	Continuous discovery	Sections 9, 24	Complete
NFR-025-002	Enterprise scalability	Sections 9, 25, 26	Complete
NFR-025-003	Low-latency inventory updates	Sections 9, 25	Complete
NFR-025-004	Explainability	Sections 9, 22, 36	Complete
NFR-025-005	Auditability	Sections 9, 27, 39	Complete
NFR-025-006	Repository stability	Sections 21, 31, 38	Complete

50. Traceability Matrix

Source Target Relationship		
--- --- ---		
IS-025 Purpose	EA-0025 Objective	Defines why the engine exists

Discovery Engine	FR-025-001	Discovers enterprise assets
Asset Inventory Engine	FR-025-002	Maintains inventory
Asset Classification Engine	FR-025-003	Classifies assets
Ownership Intelligence Engine	FR-025-004	Assigns ownership rationale
Relationship Discovery Engine	FR-025-004	Establishes relationship lineage
Lifecycle Management Engine	Asset lifecycle	Tracks asset state
Configuration Enrichment Engine	Asset intelligence	Enriches configuration data
Event Publisher	FR-025-006	Publishes asset events
Security Data Lake Integration	Discovery telemetry	Supplies history and telemetry
AI Decision Integration	Ownership recommendations	Supplies confidence and recommendations
Risk & Vulnerability Integration	Asset criticality	Supplies risk, exposure, and vulnerability context
Compliance Integration	Asset governance	Supports policy and audit
Repository Validation	Repository Standard	Confirms no top-level redesign

51. Engineering Journal

Journal Entry - EA-0025

EA-0025 was created to archive completion of IS-025 - AQELYN Cyber Asset Discovery & Inventory Intelligence Engine.

The archive records the expansion of AQELYN into authoritative cyber asset inventory intelligence. IS-025 defines the structure needed to continuously discover assets, classify assets, assign ownership, discover relationships, enrich configuration, maintain lifecycle state, govern inventory quality, and publish asset events.

The engineering design preserves the fixed AQELYN repository structure and maintains backward compatibility with previously completed engines.

Lessons Learned

Asset discovery must be modeled separately from vulnerability intelligence and attack surface management. Vulnerability intelligence identifies weaknesses, exposure management identifies reachability, and asset inventory establishes the authoritative record of what exists and how it is governed.

Governance Note

EA-0025 follows the master-document publication workflow. The Markdown file is the authoritative source, and PDF/HTML representations are generated from the same content.

52. Examples

52.1 Example Asset Record

```
asset_id: ASSET-0001
asset_type: cloud_instance
owner: infrastructure_security_team
lifecycle_state: active
classification: mission_critical
confidence: 0.94
```

52.2 Example Asset Relationship

```
relationship_id: REL-1001
source_asset: ASSET-0001
target_asset: APP-2001
relationship_type: supports_application
confidence: 0.88
```

52.3 Example Asset Classification

```
classification_id: CLASS-3001
business_criticality: high
mission_alignment: mission_alpha
regulatory_classification: security_sensitive
```

52.4 Example Asset Event

```
{
  "event_type": "asset.discovered",
  "asset_id": "ASSET-0001",
  "asset_type": "cloud_instance",
  "source_engine": "aqelyn_cyber_asset_discovery_inventory_intelligence_engine"
}
```

53. Manifest Summary

Archive contents include:

```
README.md
MD/EA-0025.md
PDF/EA-0025.pdf
HTML/index.html
HTML/styles.css
manifest/manifest.json
requirements/Requirements_Matrix.md
traceability/Traceability_Matrix.md
journal/Engineering_Journal.md
diagrams/Architecture.svg
diagrams/Component.svg
diagrams/Workflow.svg
diagrams/EventFlow.svg
diagrams/Integration.svg
examples/example_asset_inventory.md
```

54. Final Archive Statement

EA-0025 is the Engineering Archive for IS-025 - AQELYN Cyber Asset Discovery & Inventory Intelligence Engine.

It records the completed specification, the architectural decisions, the integration model, the repository impact, the risk posture, verification requirements, acceptance criteria, archive index update, and the engineering publication standard.

```
EA-0025 COMPLETE
IS-025 COMPLETE
NEXT: IS-026
```